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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (E&E E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE401/20EE401/20EC401 – PROBABILITY THEORY AND NUMERICAL METHODS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **Unit – I** **Two full** questions from **Unit – II & Unit – III**.

2) Statistical tables may be provided.

Unit – I

Marks BT* CO* PO*

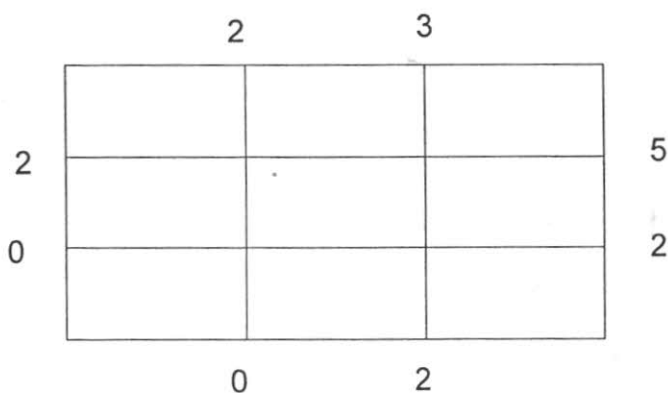
1. a) Evaluate $\int_0^6 \frac{1}{x^2+1} dx$ using Trapezoidal rule by taking seven ordinates. Compare your answer with the actual value of the integral. 7 L*2 1 2
- b) With usual notations prove that
 - i. $E = 1 + \Delta$
 - ii. $E^{-1} = 1 - \nabla$
 - iii. $\Delta = \nabla E = E \nabla$.7 L3 1 1
- c) Using Lagrange's interpolation formula find $f(4)$ given that $f(0) = 1$, $f(1) = 3$, $f(3) = 55$ and $f(5) = 251$. 6 L2 1 2
2. a) Derive Newton's forward difference interpolation formula. 7 L3 1 1
- b) Compute $y'(4)$ and $y''(4)$ given that

x	0	1	2	3	4
y	0	3	16	27	0

7 L2 1 2
- c) Evaluate $\int_0^1 \frac{1}{x+1} dx$ using Simpson's $3/8^{\text{th}}$ rule by taking seven ordinates and hence find an approximate value of $\log 2$. 6 L2 1 2

Unit – II

3. a) Using the method of false position find a real root of the equation $x^3 - 3x + 1 = 0$ in $[0, 1]$. Carry out 3 iterations. 7 L2 2 2
- b) Solve the equation $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square mesh with sides $x = 0 = y$, $x = 3 = y$ with $u = 0$ at the boundary and mesh length $h = 1$. 7 L2 3 2
- c) Using Taylor's series method find an approximate value of $y(0.1)$ given that $y' - 2y = 3e^x$, $h = 0.1$ and $y(0) = 0$. 6 L2 3 2
4. a) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown in the adjoining figure.



7 L2 3 2

P.T.O.

- b) Evaluate the pivotal values of $u_{tt} = 16u_{xx}$ up to $t = 1$ given that $u(0, t) = u(5, t) = 0, u_t(x, 0) = 0, u(x, 0) = x^2(5 - x), k = 0.5$ and $h = 1$.
- c) Using Newton-Raphson method find a real root of $x^3 - 5x + 3 = 0$ near 2. Carry out 3 iterations.
5. a) Using Milne's method find an approximate value of $y(0.4)$ given that $y' = e^{-x} - y$ and
- | | | | | |
|-----|---|--------|--------|--------|
| x | 0 | 0.1 | 0.2 | 0.3 |
| y | 0 | 0.0905 | 0.1637 | 0.2222 |
- Use corrector formula twice.
- b) Apply Runge-Kutta method of order 4 to compute an approximate value of $y(0.1)$ given that $y' = x + y^2, h = 0.1$ and $y(0) = 1$.
- c) Use Schmidt explicit method to obtain numerical solution of $u_t = u_{xx}, u(0, t) = u(1, t) = 0, u(x, 0) = \sin \pi x$ with $h = 1/3$ and $k = 1/36$ for two time levels.

7 L2 3 2

6 L2 2 2

7 L2 2 2

7 L2 3 2

6 L2 3 2

Unit – III

6. a) State and prove Bayes' theorem.
- b) The probability distribution of a random variable X is as given in the following table.

X	-2	-1	0	1	2	3
$P(X)$	0.1	k	0.2	$2k$	0.3	k

Determine k . Also find mean and variance of X .

7 L2 4 2

- c) The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured then find the probability that
- Exactly one will be defective.
 - None will be defective.
 - At least two will be defective.
7. a) With usual notations prove that $V(aX + b) = a^2V(X)$ and $V(X) = E(X^2) - [E(X)]^2$.
- b) If X is a continuous random variable with pdf $f(x) = kx(1 - x), 0 < x < 1$. Determine k . Also find cdf $F(X)$ of X .
- c) If X has a Poisson distribution and $P(X = 2) = \frac{2}{3}P(X = 1)$ then find $P(X = 0)$ and $P(X = 3)$.
8. a) If a continuous random variable X has exponential distribution with parameter, α then prove that $E(X) = 1/\alpha$ and $V(X) = 1/\alpha^2$.
- b) X and Y are random variables with joint probability distribution function given by the following table.
- | | | | |
|------------------|-----|-----|-----|
| $X \backslash Y$ | -3 | 2 | 4 |
| 1 | 0.1 | 0.2 | 0.2 |
| 3 | 0.3 | 0.1 | 0.1 |
- Find the marginal distribution of X and Y . Also find $Cov(X, Y)$.
- c) If X has distribution $N(2, 0.16)$ then find $P(X \geq 2.3)$ and $P(1.8 \leq X \leq 2.1)$

7 L3 4 1

6 L2 5 2

7 L3 4 1

7 L2 4 2

6 L2 5 2

7 L3 5 1

7 L2 5 2

6 L2 5 2